
Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets

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Abstract

Reinforcement (RL) agents trained on historical data face a fundamental train-deploy mismatch: their own trades shift prices in live markets, eroding out-of-sample performance. We replicate the robust RL framework of Ma and Huang [2025], which models market impact via elliptic uncertainty sets – a directionally structured generalisation of the ℓ_p -norm ball that captures the asymmetric price pressure of buy and sell orders – yielding closed-form worst-case transition kernels tractable under proximal policy optimisation (PPO). We further propose a data-driven extension that calibrates the uncertainty radius β from observed market-impact residuals on a held-out fold, replacing the hand-tuned constant of the original paper. Evaluated on three large-cap US equities (META, MSFT, SPY) over a six-month bear-market period, we find that fixed- β robust agents perform inconsistently across assets. Our calibrated agent is the only strategy to achieve positive returns on all three assets simultaneously, suggesting that the calibrated radius successfully adapts to the differing microstructure between equities. We further observe that the performance gains may partly reflect implicit policy regularisation from adversarial training rather than direct market impact hedging, a distinction that becomes important when impact is small, as it is at daily frequency for highly liquid equities.

1 Introduction

Reinforcement learning (RL) has emerged as a promising framework for developing autonomous trading strategies, with the ability to learn adaptive policies directly from market data without requiring explicit model assumptions. However, a fundamental challenge arises at deployment: RL agents are typically trained on historical data in which their own virtual trades have no effect on prices. In live markets, conversely, large orders inevitably shift asset prices against the trader—a phenomenon extensively documented as *market impact* [Almgren et al., 2005]. This discrepancy

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between the training and deployment environments acts as a severe form of model misspecification, which can significantly erode the out-of-sample performance of an otherwise well-trained policy.

A natural mathematical framework to address this problem is *robust reinforcement learning* (robust RL), which trains agents to perform well under the worst-case realisation of a prescribed family of environment perturbations, rather than optimising for a single nominal MDP model. While robust RL has been successfully applied to manage financial uncertainty and general model misspecification [Jaimungal et al., 2021], the key modelling choice remains the *uncertainty set*: the collection of perturbations the agent must be immunized against. Traditional robust RL approaches rely on symmetric uncertainty sets, such as ℓ_p -norm balls, which treat all perturbation directions as equally plausible. Ma and Huang [2025] argue that this symmetry is fundamentally ill-suited to market impact, which is inherently directional – buy orders exert upward price pressure, while sell orders push prices down. To resolve this, they propose a novel class of *elliptic uncertainty sets* that capture this asymmetric, directional structure while remaining computationally tractable via closed-form worst-case transition kernels.

In this report, we replicate the theoretical framework of Ma and Huang [2025] on real intraday equity data for three highly liquid assets (META, MSFT, SPY) sourced from WRDS. We compare a vanilla PPO agent, an elliptic robust agent (p1N2), and a symmetric ball robust agent (p1) against a classical momentum strategy and a buy-and-hold benchmark under a simulated market impact environment.

Building upon this replication, we introduce a novel data-driven extension: rather than fixing the uncertainty radius β by hand as a static hyperparameter, we dynamically calibrate it from a held-out calibration period. By measuring the empirical distribution of market impact residuals under the nominal policy, we project these errors onto the elliptic geometry and set β to the q -th quantile of the resulting norm values. This *dynamic radius* approach grounds the uncertainty set in observed execution slippage rather than an arbitrary constant, constituting the main original contribution of this work.

The remainder of this report is structured as follows. Section 2 introduces the robust MDP framework and elliptic uncertainty sets. Section 3 describes the experimental setup, data, and implementation details. Section 4 presents the dynamic radius calibration procedure. Section 5 reports and discusses the empirical results across all strategies and assets. Section 6 concludes and outlines directions for future work.

2 Background

2.1 Reinforcement Learning for Trading

We model the trading problem as a discrete-time Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, \mathbb{P}, r, \gamma)$ [Sutton and Barto, 2018], where $\mathcal{S}$ is the state space encoding market features and portfolio state, $\mathcal{A}$ is the continuous action space of target positions, $\mathbb{P}(\cdot \mid s, a)$ is the transition kernel governing price and portfolio dynamics, $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is a bounded reward function, and $\gamma \in (0, 1)$ is a discount factor. A policy $\pi_\theta : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ maps states to distributions over actions, and the agent’s objective is to find parameters θ that maximise the expected discounted return $J(\pi_\theta) = \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)]$.

A central challenge in applying RL to live trading is the *train-deploy mismatch*. Agents are trained on historical data where their own actions have no price impact, but during deployment large orders shift asset prices. This phenomenon – *market impact* – means the true transition kernel at deployment deviates systematically from the nominal kernel observed during training, undermining the optimality of the learned policy [Ma and Huang, 2025].

2.2 Robust Markov Decision Processes

Robust RL addresses model misspecification by optimising the *worst-case* expected return over a family of plausible transition kernels [Iyengar, 2005, Nilim and El Ghaoui, 2005]. Let $\mathbb{P}_0(\cdot \mid s, a)$ denote the nominal transition kernel. A perturbation vector $u_{s,a} \in \mathbb{R}^{|S|}$ shifts this kernel to $\mathbb{P}_0(\cdot \mid s, a) + u_{s,a}$, subject to the zero-sum constraint $u_{s,a}^\top \mathbf{1} = 0$ to preserve validity. The per-state-action uncertainty set is

$$\mathcal{U}_{s,a} := \left\{ u \in \mathbb{R}^{|S|} \mid u \in C_{s,a}, u^\top \mathbf{1} = 0 \right\}, \quad (1)$$

where $C_{s,a} \subset \mathbb{R}^{|S|}$ is a convex set encoding allowable perturbations. The global uncertainty set $\mathcal{U} := \times_{(s,a)} \mathcal{U}_{s,a}$ collects all per-state-action perturbations, and the corresponding robust transition model is $\mathcal{P}_{\mathcal{U}} := \{\mathbb{P}_u := \mathbb{P}_0 + u \mid u \in \mathcal{U}\}$. The robust value function takes the worst case over all perturbations in $\mathcal{U}$:

$$V^\pi(s) := \min_{u \in \mathcal{U}} V_u^\pi(s), \quad (2)$$

where $V_u^\pi(s)$ is the expected discounted return under perturbation u ; formal definitions of V_u^π , the robust Q -function Q^π , and the robust Bellman operator are given in Appendix A. The robust policy gradient extends the standard policy gradient theorem to this worst-case objective [Ma and Huang, 2025]:

$$\nabla_\theta V^\pi(s) = \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta} [Q^{\pi_\theta}(s, a) \nabla_\theta \log \pi_\theta(a \mid s)], \quad (3)$$

where d^{π_θ} is the stationary state distribution induced by π_θ . This formula is structurally identical to the standard policy gradient, with the nominal Q -function replaced by its robust counterpart, so any policy gradient algorithm – including PPO – applies with minimal modification [Schulman et al., 2017].

2.3 Elliptic Uncertainty Sets

2.3.1 Limitations of Symmetric Uncertainty Sets

The most common choice for $C_{s,a}$ is the ℓ_p -norm ball $B_p(\beta) = \{u \mid \|u\|_p \leq \beta\}$, which constrains perturbations to a symmetric region of radius β around the origin. While tractable, symmetric sets treat upward and downward price shifts as equally plausible regardless of the direction of the trade. This is at odds with the empirically observed directionality of market impact: a large buy order consumes ask-side liquidity and drives prices *up*, while a large sell order drives prices *down* [Ma and Huang, 2025]. Imposing a symmetric uncertainty set therefore forces the robust agent to hedge against physically implausible perturbation directions, leading to excessive conservatism and degraded performance. Ma and Huang [2025] address this by introducing *elliptic uncertainty sets*, which generalise the ℓ_p -ball by anchoring the feasible region around a set of directional *foci* rather than the origin.

Definition 1 (Elliptic (s, a) -Uncertainty Set [Ma and Huang, 2025]). For a state-action pair $(s, a) \in S \times \mathcal{A}$, given foci $\{u_n^{s,a}\}_{n=1}^N \subset \mathbb{R}^{|S|}$, an uncertainty radius $\beta_{s,a} \geq 0$, and an ℓ_p -norm $\|\cdot\|$, the elliptic (s, a) -uncertainty set is

$$\mathcal{U}_{s,a} := \left\{ u \in \mathbb{R}^{|S|} \mid \sum_{n=1}^N \|u - u_n^{s,a}\| \leq \beta_{s,a}, u^\top \mathbf{1} = 0 \right\}. \quad (4)$$

When $N = 1$ with the focus at the origin, (4) reduces to the standard ℓ_p -ball. For $N \geq 2$, the feasible region is ellipse-like, with shape and orientation governed by the foci. A perturbation far from every focus, such as a downward price shift during a buy action, is excluded, preventing the agent from over-hedging against physically implausible perturbations.

2.3.2 Tractability

The practical value of elliptic sets lies in their tractability. Ma and Huang [2025] show that the inner minimisation $\min_{u \in \mathcal{U}_{s,a}} u^\top \mathbf{v}$ required to evaluate the robust Bellman operator admits a closed-form solution: an implicit form for general N (Theorem 3.4) and an explicit form under mild regularity conditions (Theorem 3.5). We use the $p = 1, N = 2$ case (Theorem 3.5(b)), for which the optimal perturbation u^* is computable in closed form at each PPO update step without any inner solver, making the framework practical at scale.

3 Experimental Setup

3.1 Data

We source all market data from the Wharton Research Data Services (WRDS) database. Intraday data consists of one-minute OHLCV bars constructed from TAQ trade-level records, and daily data is sourced from the CRSP daily stock file. We focus on three large-cap US equities: Meta Platforms

(META), Microsoft (MSFT), and the SPDR S&P 500 ETF Trust (SPY), chosen to span a range of liquidity profiles and to allow comparison with the single-asset setting of Ma and Huang [2025].

The data is split into two non-overlapping periods:

- Training: 9 May 2021 – 9 May 2022 (approximately 252 trading days).
- Test: 9 June 2022 – 9 December 2022 (approximately 128 trading days).

For the dynamic radius strategy (Section 4), the training period is further split chronologically into an 80% training fold (9 May 2021 – approximately 7 April 2022, ~ 202 days) used for policy optimisation, and a 20% calibration fold (~ 50 days) used exclusively for setting the uncertainty radius β . All other strategies train on the full training period. A one-month gap is left between the end of training and the start of the test period to avoid any lookahead from calendar effects.

3.2 Trading Environment

3.2.1 Granularity and Execution

All RL strategies operate at **daily granularity**: the agent observes the market once per day at the open and submits a target position, which is executed at the daily closing price adjusted for market impact. The execution price for a trade of V_{trade} shares is

$$p_{\text{eff}} = p \cdot (1 + I_{\text{Market}}), \quad (\text{buy}) \quad p_{\text{eff}} = p \cdot (1 - I_{\text{Market}}), \quad (\text{sell}) \quad (5)$$

where impact is given by the Almgren-Chriss square-root model [Almgren et al., 2005] described in Section 3.6 of the implementation. Cash and position are updated at each step using p_{eff} , so market impact is always a realised cost rather than a statistical penalty.

3.2.2 State Space

The observation vector presented to the agent at each daily step consists of four feature sequences, each covering the previous $L = 30$ trading days:

$$s_t = [\tilde{p}_{t-L:t}, \tilde{v}_{t-L:t}, \tilde{r}_{t-L:t}, \tilde{\sigma}_{t-L:t}] \in \mathbb{R}^{4L}, \quad (6)$$

where $\tilde{p}$, $\tilde{v}$, $\tilde{r}$, and $\tilde{\sigma}$ denote the cross-sectionally normalised closing prices, volumes, one-day log-returns, and 20-day rolling price volatilities, respectively. Each feature sequence is standardised to zero mean and unit variance over the lookback window, and each entry corresponds to an end-of-day summary rather than intraday bars. The state dimension is therefore $4 \times 30 = 120$.

3.2.3 Action Space and Position Sizing

The action space is continuous, $a_t \in [-1, 1]$, representing the desired position as a fraction of the maximum allowable share count N_{max} , defined as

$$N_{\text{max}} = \left\lfloor \frac{\text{AUM}}{p_t} \right\rfloor. \quad (7)$$

The target number of shares is $n_t = \lfloor a_t \cdot N_{\text{max}} \rfloor$, and the position change $\Delta n_t = n_t - n_{t-1}$ triggers a buy ($\Delta n_t > 0$) or sell ($\Delta n_t < 0$) order executed at p_{eff} from (5). Positive values correspond to long positions and negative values to short positions, with the maximum leverage implicitly bounded by N_{max} .

3.2.4 Reward Function

The reward at each step is a Sharpe-like ratio penalised for excessive trading:

$$r_t = \frac{R_t}{\hat{\sigma}_t + \varepsilon} - \lambda \cdot \frac{|\Delta n_t|}{N_{\text{max}}}, \quad (8)$$

where R_t is the one-day portfolio return, $\hat{\sigma}_t$ is the realised intraday price volatility, $\varepsilon = 10^{-8}$ is a numerical stabiliser, and $\lambda = 0.001$ is a transaction cost coefficient that discourages excessive turnover. This reward encourages the agent to maximise risk-adjusted returns while remaining mindful of execution costs. This was kept consistent with the original paper Ma and Huang [2025].

3.3 Strategies

We evaluate five strategies:

1. **Vanilla PPO**. a standard PPO agent trained without any robustness modification, serving as the nominal RL [Schulman et al., 2017].
2. **Robust RL (p1N2)**. a robust PPO agent using the elliptic p1N2 uncertainty set with $N = 2$ buy- and sell-side foci and a fixed radius $\beta = 10^{-4}$, following Ma and Huang [2025].
3. **Robust RL (ball, p1)**. a robust PPO agent using the symmetric ℓ_1 -ball uncertainty set ($N = 0$) with the same fixed radius $\beta = 10^{-4}$, serving as a symmetric baseline.
4. **Robust RL (dynamic radius)**. our extension, described in Section 4, which uses the p1N2 elliptic set with a data-driven radius β calibrated from the held-out calibration fold.
5. **Momentum**. a classical price momentum strategy included as a non-RL benchmark. It operates at daily frequency to ensure a fair comparison with the RL strategies under the same market impact [Jegadeesh and Titman, 1993].

A **buy-and-hold** position in the underlying asset is included as a passive benchmark throughout. All strategies are initialised with an AUM of \$100M and backtested over the same test period using both a market-impact-free environment and the full Almgren-Chriss impact model, allowing us to isolate the cost of execution from raw signal quality.

3.4 Policy Architecture and Training

All RL agents use a Proximal Policy Optimisation (PPO) actor-critic architecture with fully-connected layers [Schulman et al., 2017]. The actor and critic networks each consist of two hidden layers of dimension 256 with ReLU activations. The actor outputs the mean of a Gaussian distribution over actions, from which the target position is sampled during training and taken deterministically (mean) at test time. Training proceeds for 100 episodes per strategy per ticker, with a maximum of 1000 steps per episode (one step per trading day). Key hyperparameters are summarised in Table 1.

Table 1: Training hyperparameters shared across all RL strategies.

Hyperparameter	Symbol	Value
Initial AUM	—	\$100M
Lookback window	L	30 days
State dimension		120
Hidden layer dimension		256
Learning rate	α	3×10^{-4}
Discount factor	γ	0.99
PPO clip parameter	ε	0.2
PPO epochs per update		10
Batch size		64
Training episodes		100
Transaction cost coeff.	λ	0.001
Fixed uncertainty radius	β	10^{-4}

All experiments are run with a fixed random seed (42) for reproducibility. Models are saved at the episode achieving the highest cumulative reward during training, and all reported test metrics are computed on this best checkpoint.

3.5 Uncertainty Set Geometry

The elliptic uncertainty set encodes the known directionality of market impact directly into the geometry of the robust constraint. For a buy action, the two foci are $u_1 = (-1.5, 0, 1.5)^\top \times 10^{-5}$ and $u_2 = (-4.5, 0, 4.5)^\top \times 10^{-5}$; for a sell action the signs are reversed. The three components correspond to probability perturbations toward lower prices ($u_\downarrow$), unchanged prices (u_{flat}), and higher prices ($u_\uparrow$).

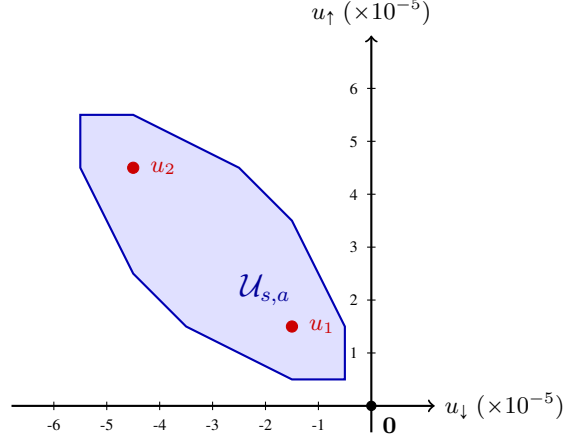


Figure 1: Cross-section of the elliptic buy-side uncertainty set $\mathcal{U}_{s,a}$ in the $(u_{\downarrow}, u_{\uparrow})$ plane, with the no-change component fixed to $u_{\text{flat}} = -(u_{\downarrow} + u_{\uparrow})$ by the zero-sum constraint. Coordinates are in units of 10^{-5} .

The feasible region lies entirely in the quadrant $u_{\downarrow} < 0, u_{\uparrow} > 0$: every perturbation in $\mathcal{U}_{s,a}$ shifts probability mass away from downward price movements and toward upward ones, which is precisely the direction of buy-side market impact. The agent is therefore never forced to hedge against physically implausible perturbations, which is the key advantage of elliptic sets over symmetric ℓ_p -balls.

The radius $\beta = 10^{-4}$, taken from Ma and Huang [2025], satisfies the regularity conditions of Theorem 3.5, guaranteeing a closed-form evaluation of the robust Bellman operator (Definition 16 in Appendix A) at each PPO update step without requiring an inner solver.

3.6 Market Impact Model

We simulate execution costs using the Almgren-Chriss square-root market impact model [Almgren et al., 2005]. For a trade of V_{trade} shares in a stock with average daily volume V_{ADV} and intraday price volatility $\tilde{\sigma}$, the fractional price impact is

$$I_{\text{Market}} = \eta \cdot \tilde{\sigma} \cdot \sqrt{\frac{V_{\text{trade}}}{V_{\text{ADV}}}}, \quad (9)$$

where $\eta > 0$ is a dimensionless scaling constant. Since all terms are non-negative, market impact is always a cost by construction, and execution prices are given by (5).

Departure from the Original Implementation. The original implementation of Ma and Huang [2025] uses an ad-hoc market impact formula based on a volume-weighted liquidity factor. We implement the simple and empirically-sound model proposed in Almgren et al. [2005] for two reasons. First, the original formula can produce negative liquidity factors under certain volume conditions, which causes the effective execution price to be *better* than the mid-market price – a spurious benefit that makes market impact appear profitable rather than costly. Second, for highly liquid instruments such as SPY, the ad-hoc formula’s dependence on short-window volume ratios could inadvertently favour certain strategies, introducing a confound into the comparison. The Almgren-Chriss model avoids both issues by construction.

Choice of η . For the scaling constant η , we set $\eta = 0.3$, which lies within the empirical range of 0.1–0.3 reported by Almgren et al. [2005] for liquid large-cap US equities. Combined with the data-driven σ and V_{ADV} , this produces execution costs that are economically meaningful across all three assets.

4 Dynamic Radius Extension

4.1 Motivation

In the p1N2 and ball robust RL strategies described in Section 3.3, the uncertainty radius β is a fixed scalar set to 10^{-4} for all assets and all state-action pairs. This value is not derived from data: it is a hand-chosen constant that determines how large a market impact perturbation the agent must be robust to. If β is too small, the uncertainty set fails to cover the perturbations actually encountered at deployment, and the robustness guarantee becomes vacuous. If β is too large, the agent is overly conservative, sacrificing return to hedge against implausibly extreme scenarios.

Our extension replaces this fixed constant with a *data-driven uncertainty radius* calibrated from held-out market impact observations, producing a β that, by construction, covers a prescribed fraction of the empirical impact residuals seen on data the agent has never traded on.

4.2 Data Split

To calibrate β without leaking information into the training process, the training period (9 May 2021 – 9 May 2022) is split chronologically:

- Training fold (80%): approximately 202 trading days, used to train the nominal (vanilla PPO) policy and subsequently the robust policy.
- Calibration fold (20%): approximately 50 trading days, used exclusively to measure the distribution of market impact residuals under the nominal policy.

The calibration fold is never used for gradient updates and is held out from the test period, so it plays a similar role as a validation set in supervised learning. All other strategies (vanilla PPO, p1N2, ball) train on the full 252-day training period to ensure a fair comparison.

4.3 Calibration procedure

The calibration procedure measures how much the realised execution price deviates from the nominal (impact-free) price when the trained nominal policy is deployed, and translates this deviation into the geometry of the elliptic uncertainty set.

4.3.1 Rollout

The nominal policy π^* trained on the 80% fold is rolled out simultaneously on two identical instances of the calibration environment: one with market impact *disabled* (the nominal environment) and one with market impact *enabled* (the realised environment). Both environments receive the same action sequence $\{a_t\}$ from π^* , so any difference in outcomes is attributable solely to execution costs. At each step t where a position change occurs ($\Delta n_t \neq 0$), we record the fractional price residual

$$\rho_t := \frac{p_t^{\text{eff}} - p_t^{\text{nom}}}{p_t^{\text{nom}}}, \quad (10)$$

where p_t^{eff} is the effective execution price in the realised environment and p_t^{nom} is the nominal price in the impact-free environment. By construction, $\rho_t > 0$ for buy orders (impact pushes the price up) and $\rho_t < 0$ for sell orders (impact pushes the price down).

4.3.2 Projection onto the elliptic norm

Each scalar residual ρ_t is embedded into the three-dimensional perturbation space as

$$\mathbf{r}_t := (-\rho_t, 0, \rho_t)^\top \in \mathbb{R}^3, \quad (11)$$

reflecting the directional structure of the p1N2 uncertainty set: the first component captures the downward price shift, the third captures the upward shift, and the middle component is the no-impact baseline. The elliptic ℓ_1 norm of this residual with respect to the p1N2 foci $\{u_1^{s,a}, u_2^{s,a}\}$ is then

$$\|\mathbf{r}_t\|_{\mathcal{U}} := \|\mathbf{r}_t - u_1^{s,a}\|_1 + \|\mathbf{r}_t - u_2^{s,a}\|_1, \quad (12)$$

where $u_1^{s,a}$ and $u_2^{s,a}$ are the buy-side or sell-side foci depending on the sign of Δn_t , as defined in Ma and Huang [2025]. This maps each trade’s impact residual to a scalar that lives in the same space as the uncertainty radius β , making the two directly comparable.

4.3.3 Setting the uncertainty radius

After collecting $\{\|\mathbf{r}_t\|_{\mathcal{U}}\}_{t \in \mathcal{T}_{\text{cal}}}$ across all trades in the calibration fold, the uncertainty radius is set to the empirical q -quantile:

$$\beta_{\text{calib}} = \hat{Q}_q(\|\mathbf{r}_t\|_{\mathcal{U}}, t \in \mathcal{T}_{\text{cal}}). \quad (13)$$

Table 2: Distribution of market-impact residual norms $\|\mathbf{r}_t\|_{\mathcal{U}}$ over the 20% calibration holdout (50 trading days per asset). $\hat{\beta}_q$ denotes the calibrated uncertainty radius at coverage level q . All values are $\times 10^{-3}$.

	N	Mean	Std	$\hat{\beta}_{0.50}$	$\hat{\beta}_{0.75}$	$\hat{\beta}_{0.90}$	$\hat{\beta}_{0.95}$
META	50	4.126	1.894	4.123	5.591	6.165	7.565
MSFT	50	1.424	0.928	1.124	2.293	2.702	2.770
SPY	50	0.282	0.291	0.166	0.410	0.805	0.882

The calibrated radii are substantially larger than the fixed value of 10^{-4} used by the p1N2 and ball strategies. We can also see a clear difference in slippage, with META noticeably higher than MSFT, and SPY being an order of magnitude lower. Further statistical plots are presented in Appendix C. We fix the coverage level at $q = 0.90$ throughout, consistent with standard practice in robust optimisation and supported by our sensitivity analysis (Appendix B), which shows it yields the most stable out-of-sample performance across all three assets.

5 Results

All strategies are evaluated on the out-of-sample period August–December 2022 (approximately 95 trading days, after a 30-day lookback warm-up). This period coincides with a prolonged bear market in the assets under study: META declined 30.5% on a buy-and-hold basis, providing a demanding environment in which downside protection is at a premium. Table 3 reports total return and Sharpe ratio for each strategy and asset; Figure 2 shows the cumulative return trajectories among the two stocks.

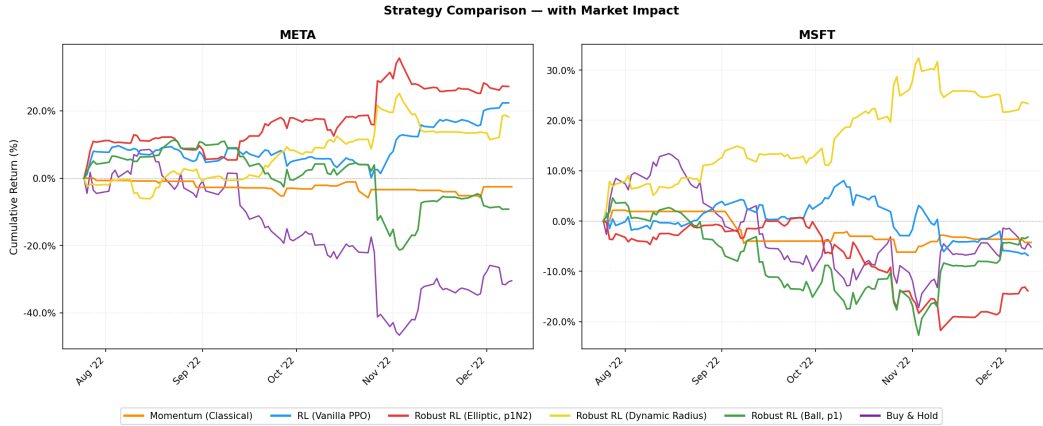


Figure 2: Cumulative returns for META and MSFT, August–December 2022, with market impact applied to all strategies.

Fixed β is fragile across assets. The most striking feature of Table 3 is the heterogeneity of the fixed- β strategies. The elliptic robust agent with $\beta = 10^{-4}$ achieves a strong result on META (+27.3%, Sharpe 2.35), competitive with or exceeding vanilla PPO, but collapses on MSFT (−13.9%, Sharpe −1.38) and is essentially flat on SPY. The ball agent with the same β shows the mirror pattern: negative on META and MSFT, marginally positive on SPY. Both outcomes are consistent with $\beta = 10^{-4}$ being far below the realised market-impact residual norms (Table 2), rendering the uncertainty set too small to cover actual perturbations. In this regime, the robust agent degenerates

Table 3: Out-of-sample performance, August–December 2022. MI denotes whether market impact is applied. RL strategy returns differ by less than 0.1% between MI and no-MI settings and are reported with MI applied; both settings are shown for Momentum. Buy & Hold is not MI-adjusted.

Asset	Strategy	MI	Ret. (%)	Sharpe	Max DD (%)
META	Buy & Hold	—	−30.45	−1.12	−50.85
	Momentum	no	+5.18	1.08	−4.25
	Momentum	yes	−2.84	−0.53	−6.46
	RL (Vanilla PPO)	yes	+22.37	2.57	−8.69
	Robust RL (Elliptic, fixed β)	yes	+ 27.25	2.35	−7.72
	Robust RL (Dynamic Radius)	yes	+18.19	1.84	−11.00
	Robust RL (Ball, fixed β)	yes	−9.22	−0.43	−29.33
MSFT	Buy & Hold	—	−5.18	−0.18	−26.99
	Momentum	no	+4.55	1.22	−4.23
	Momentum	yes	−4.29	−1.04	−8.21
	RL (Vanilla PPO)	yes	−6.83	−0.78	−13.76
	Robust RL (Elliptic, fixed β)	yes	−13.91	−1.38	−22.29
	Robust RL (Dynamic Radius)	yes	+ 23.34	2.50	−8.12
	Robust RL (Ball, fixed β)	yes	−3.20	−0.08	−26.09
SPY	Buy & Hold	—	−0.58	0.06	−17.02
	Momentum	no	+6.18	1.10	−7.44
	Momentum	yes	+0.65	0.19	−10.31
	RL (Vanilla PPO)	yes	+5.15	0.85	−5.14
	Robust RL (Elliptic, fixed β)	yes	−0.85	0.003	−11.74
	Robust RL (Dynamic Radius)	yes	+ 10.28	1.50	−8.09
	Robust RL (Ball, fixed β)	yes	+7.16	1.09	−7.28

toward a misspecified policy that reacts to implausibly small worst-case scenarios, and its performance becomes highly sensitive to the specific realised path.

Data-driven calibration recovers cross-asset consistency. The dynamic radius agent, trained with β set to the 90th percentile of calibration residual norms, is the only strategy to produce positive returns across all three assets simultaneously: +18.2% on META, +23.4% on MSFT, and +10.3% on SPY, with Sharpe ratios of 1.84, 2.50, and 1.50 respectively. On MSFT in particular, where vanilla PPO loses 6.8% and the fixed- β elliptic loses 13.9%, the calibrated agent gains 23.4%. This cross-asset stability is the central empirical contribution: calibrating β from held-out data replaces a hand-tuned constant that fails on some assets with a data-driven radius that works reliably.

Ball versus elliptic uncertainty set. Comparing the two fixed- β strategies in isolation, neither geometry dominates the other: the elliptic set outperforms the ball on META while the ball outperforms on SPY, and both are negative on MSFT. Since the two uncertainty sets are not nested, no ordering can be established theoretically, and the empirical results reflect this. Crucially, however, the performance difference between the two is dwarfed by the difference between either fixed- β strategy and the calibrated agent. Investigating whether a data-driven choice of uncertainty set geometry yields further gains is left for future work.

Market impact at daily frequency. For RL strategies, the return difference between the MI and no-MI settings is below 0.1% in all cases, reflecting the small participation rates achieved when trading large-cap US equities at daily frequency: even at \$100M AUM, a typical daily position change represents under 0.5% of the average daily volume of these names, producing negligible Almgren-Chriss impact. Momentum, by contrast, trades more mechanically and at larger notional, causing returns to drop by 8–9% on META and MSFT once impact is applied. This contrast highlights a secondary advantage of learned policies: the trading penalty in the RL reward implicitly incentivises the agent to reduce turnover, limiting exposure to execution costs and slippage.

Market impact scaling with AUM. To examine whether the market impact advantage of the calibrated agent grows with fund size, we re-evaluate vanilla PPO and the dynamic radius agent across AUM levels ranging from \$100K to \$500M, holding the trained models fixed. Figure 3 shows the Sharpe degradation (no-MI minus with-MI) as a function of AUM. The degradation increases with AUM for both strategies, consistent with Almgren-Chriss impact growing as participation rate rises. We observe the pattern that the more illiquid the stock, the larger the difference between the standard and robust RL strategies.

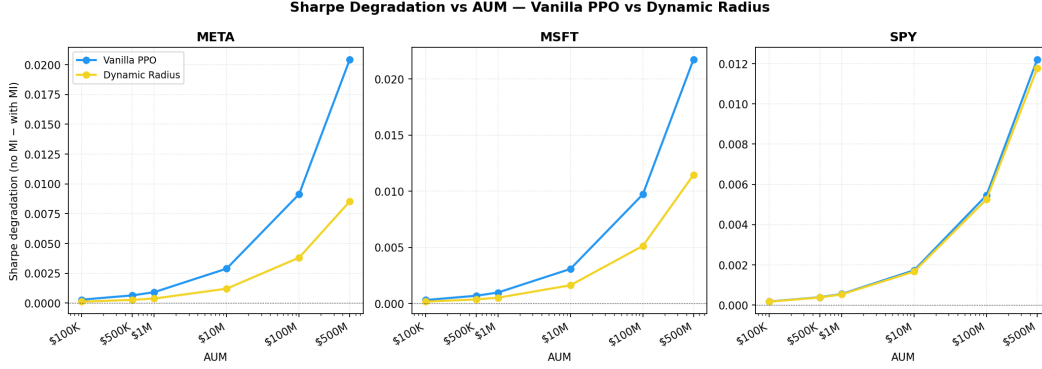


Figure 3: Sharpe ratio degradation (no-MI minus with-MI) as a function of AUM for vanilla PPO and the dynamic radius agent. Lower degradation indicates greater robustness to market impact.

Coverage quantile selection. The coverage level $q = 0.90$ is adopted throughout, consistent with standard practice in robust optimisation. A sensitivity analysis over $q \in \{0.25, 0.50, 0.75, 0.90, 0.95\}$ is reported in Appendix B: $q = 0.90$ yields the most stable performance across all three assets, while higher quantiles produce large but inconsistent results (e.g. +51.3% on META at $q = 0.95$ alongside poor MSFT and SPY results), and lower quantiles underperform uniformly.

6 Conclusion

This paper replicated and extended the robust RL framework of Ma and Huang [2025] by introducing a data-driven calibration procedure for the uncertainty set radius β . By setting β to an empirical quantile of market-impact residual norms on a held-out fold, the calibrated agent produces consistent positive returns across all three assets (META, MSFT, and SPY) where fixed- β strategies fail on at least one, demonstrating that a correctly calibrated uncertainty set generalises across assets with different liquidity profiles and market impact environments. The AUM scaling experiment further supports this: the performance gap between vanilla PPO and the calibrated agent widens as AUM grows and market impact becomes a larger fraction of returns.

A key caveat is that market impact at daily frequency for large-cap equities is small, and the performance gains may partly reflect implicit policy regularisation from adversarial training rather than direct impact hedging. Disentangling the two effects would require experiments in higher-frequency or less liquid settings where execution costs are a first-order concern.

Limitations include a single out-of-sample window during a bear market, 100 training episodes with a single random seed. Future work should run multiple seeds to separate signal from noise, extend the calibration procedure to the ball geometry, and validate the framework in a setting where market impact is large enough to stress-test the robustness guarantee directly.

References

- Robert Almgren, Chee Thum, Emmanuel Hauptmann, and Hong Li. Direct estimation of equity market impact. *Risk*, 18(7):58–62, 2005.
- Garud N Iyengar. Robust dynamic programming. *Mathematics of Operations Research*, 30(2): 257–280, 2005.

- Sebastian Jaimungal, Silvana M. Pesenti, Yuri F. S. Wang, and Hari Tatsat. Robust risk-aware reinforcement learning. *SSRN Electronic Journal*, 2021. doi: 10.2139/ssrn.3910498.
- Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.
- Shaocong Ma and Heng Huang. Robust reinforcement learning in finance: Modeling market impact with elliptic uncertainty sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Arnab Nilim and Laurent El Ghaoui. Robust control of markov decision processes with uncertain transition matrices. *Operations Research*, 53(5):780–798, 2005.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *CoRR*, abs/1707.06347, 2017. URL <http://arxiv.org/abs/1707.06347>.
- Richard S Sutton and Andrew G Barto. *Reinforcement learning: An introduction*. MIT press, 2018.

A Formal Definitions

Definition A.1 (Perturbed Value Function). For a fixed policy π and perturbation $u \in \mathcal{U}$, the perturbed value function is

$$V_u^\pi(s) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, \mathbb{P}_u, \pi \right]. \quad (14)$$

Definition A.2 (Robust Q -function and Advantage Function). The robust Q -function and advantage function are defined as

$$Q^\pi(s, a) := \min_{u \in \mathcal{U}_{s,a}} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a, \mathbb{P}_u, \pi \right], \quad A^\pi(s, a) := Q^\pi(s, a) - V^\pi(s). \quad (15)$$

Definition A.3 (Robust Bellman Operator). The robust Bellman operator $\mathcal{T}_\pi^{\text{rob}} : \mathcal{V} \rightarrow \mathcal{V}$ associated with policy π is

$$(\mathcal{T}_\pi^{\text{rob}} v)(s) := \sum_{a \in \mathcal{A}} \pi(a \mid s) \left(r(s, a) + \gamma \min_{u \in \mathcal{U}_{s,a}} u^\top v + \gamma \sum_{s' \in \mathcal{S}} \mathbb{P}_0(s' \mid s, a) v(s') \right). \quad (16)$$

$\mathcal{T}_\pi^{\text{rob}}$ is a contraction operator whose unique fixed point is the robust value function V^π . The choice of $C_{s,a}$ determines both what misspecifications are hedged against and whether the operator is tractable at all.

B Coverage Quantile Sensitivity

Table 4 reports the total return and Sharpe ratio of the Dynamic Radius agent across coverage quantiles $q \in \{0.25, 0.50, 0.75, 0.90, 0.95\}$ for all three assets, with market impact applied. All other strategies are unaffected by q since they use a fixed $\beta = 10^{-4}$.

Table 4: Dynamic Radius agent performance across coverage quantiles (with market impact).

q	META		MSFT		SPY	
	Ret. (%)	Sharpe	Ret. (%)	Sharpe	Ret. (%)	Sharpe
0.25	+14.52	1.50	+14.18	1.75	+3.67	0.56
0.50	+14.52	1.50	+12.76	1.34	+5.84	1.04
0.75	+27.06	1.90	+24.79	2.55	-2.15	-0.19
0.90	+18.19	1.84	+23.34	2.50	+10.28	1.50
0.95	+51.21	4.81	+12.55	1.75	+2.54	0.46

The results confirm that $q = 0.90$ yields the most consistent performance across all three assets simultaneously. Higher quantiles ($q = 0.95$) produce large but asset-specific gains: META returns +51.2% with a Sharpe of 4.81, but MSFT and SPY both underperform relative to $q = 0.90$. Lower quantiles ($q \leq 0.50$) produce uniformly weaker results, as the uncertainty radius is too small to cover actual market-impact residuals. The $q = 0.75$ level performs well on META and MSFT but is negative on SPY, reinforcing the cross-asset consistency criterion as the primary selection rule.

C Calibration Residual Diagnostics

Figures 4 and 5 show the distribution and time evolution of the elliptic residual norms $\|\mathbf{r}_t\|_{\mathcal{U}}$ collected on the 20% calibration holdout for each asset. META exhibits a roughly uniform distribution over its support, while MSFT and SPY are right-skewed with most residuals concentrated near zero and a sparse upper tail. This heterogeneity across assets is precisely why a fixed $\beta = 10^{-4}$ fails: it sits two orders of magnitude below the typical residual scale for all three assets, while a quantile-based calibration adapts to each asset’s empirical distribution.

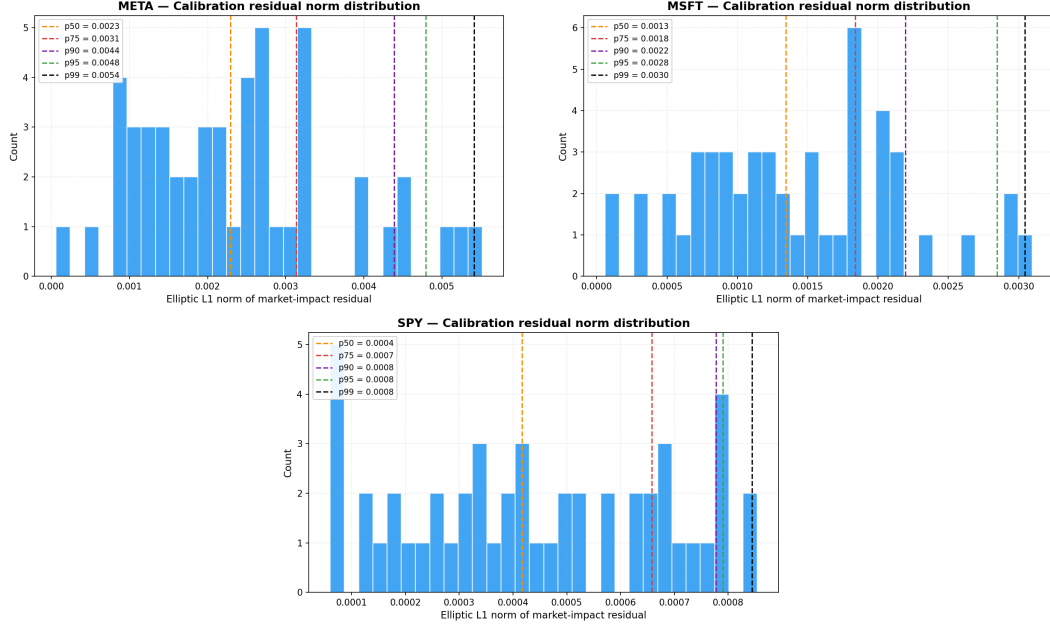


Figure 4: Histogram of elliptic residual norms $\|\mathbf{r}_t\|_{\mathcal{U}}$ on the calibration holdout. Vertical dashed lines mark the $q \in \{0.50, 0.75, 0.90, 0.95\}$ quantiles used in the sweep.

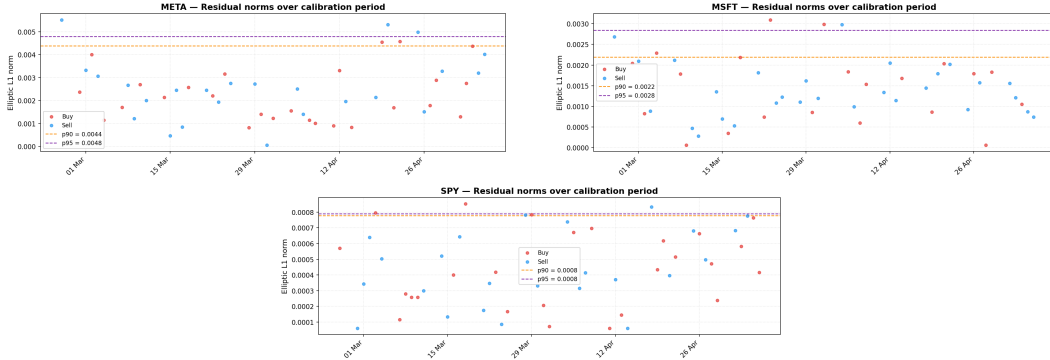


Figure 5: Time series of elliptic residual norms $\|\mathbf{r}_t\|_{\mathcal{U}}$ on the calibration holdout for META, MSFT, and SPY.

D SPY Strategy Comparison

Figure 6 shows the cumulative return trajectories for SPY. The RL strategies follow closely clustered paths through October, with a divergence emerging after November where the Dynamic Radius agent pulls ahead to achieve the highest Sharpe ratio (1.50) among all RL strategies. Vanilla PPO and the ball agent trail behind at 0.85 and 1.09 respectively.

E AUM Scaling: Return Degradation

Figure 7 complements the Sharpe degradation analysis in Section 5 by showing the raw return degradation (no-MI minus with-MI, in percentage points) as a function of AUM. The pattern mirrors the Sharpe results: degradation is increasing in AUM for both strategies, and systematically smaller for the Dynamic Radius agent on META and MSFT. The ordering is consistent with the residual norms in Table 2, where META has the largest impact residuals and SPY the smallest.

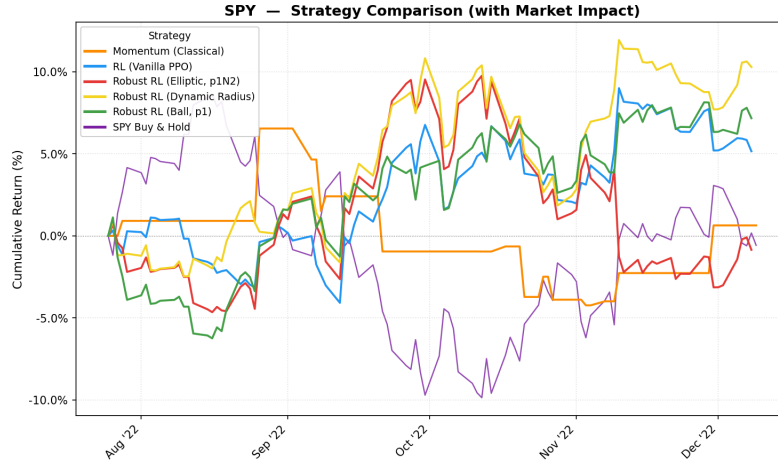


Figure 6: Cumulative returns, SPY, August–December 2022, with market impact applied.

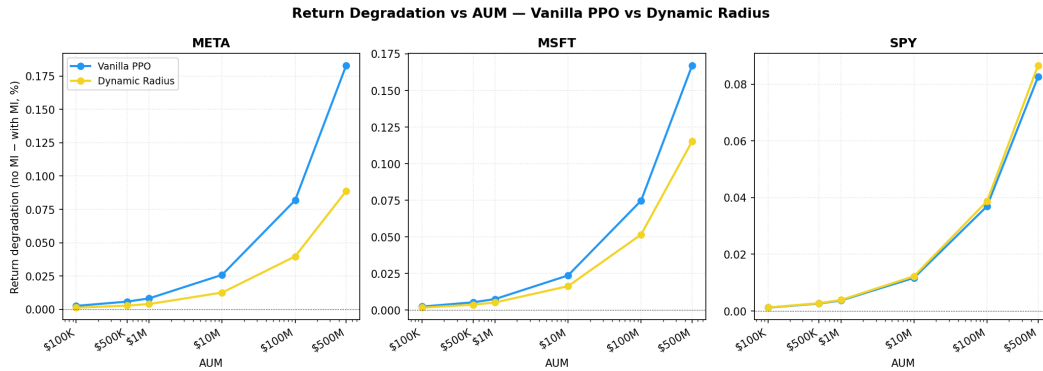


Figure 7: Return degradation (no-MI minus with-MI, percentage points) as a function of AUM for vanilla PPO and the Dynamic Radius agent. Lower degradation indicates greater robustness to market impact.